

Miniaturized Triple-Wideband Double-Overlapped e-Shaped Antenna with Parasitic Elements for High-Speed THz Wireless Indoor Communications

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Abstract— A triple-wideband miniaturized ($300 \times 300 \times 15 \mu\text{m}^3$) double-overlapped e-shaped THz antenna with parasitic elements and a non-central slotted narrow partial ground plane is proposed for high-speed indoor THz communication applications. The antenna operates efficiently across three wide frequency bands, offering $S_{11} \leq -10$ dB impedance bandwidths of 80 GHz (0.27–0.35 THz), 460 GHz (0.89–1.35 THz), and 1690 GHz (1.79–3.48 THz). Constructed using Rogers RT5880 substrate and copper conductors, the design ensures high radiation efficiency (90–92.8%), acceptable gain, good VSWR, and desirable radiation patterns. The antenna's evolutionary design process has also been thoroughly analyzed. Based on its strong performance, this antenna is well-suited for a wide range of high-speed THz indoor wireless communication systems.

Index Terms— THz antenna; Miniaturization; Triple wideband; High speed; Indoor communication.

I. INTRODUCTION

Recent advances in wireless technology require ultra-wide bandwidth to support high data rates, channel capacity, and seamless connectivity. To meet these demands, researchers are turning to the THz band (0.1–10 THz, wavelength: 3 mm–30 μm), recognized by the ITU as a key area for next-generation high-speed, secure wireless communications [1]. Ultra-dense networks are crucial for high-throughput demands in crowded areas like cities and industrial zones. The THz band offers abundant spectrum and ultra-wide bandwidth, making it ideal for small cells with limited coverage and strong LoS paths, enabling Tbps data rates. These cells can support ultra-HD video, VR, and holographic communication beyond 5G capabilities. A proof-of-concept by T. Harter et al. demonstrated 115 Gbit/s at 0.3 THz over 110 meters [2]. Meeting 40–100 Gbps for indoor and 100 Gbps for outdoor data rates remains a major challenge. While THz technology is viable for both, its limited range and penetration make it better suited for indoor use, ideal for

secure, high-speed WLAN systems within confined spaces [3]. THz communications enable direct Tbps device-to-device (D2D) links in close-range indoor environments like homes and offices, with abundant bandwidth, compact circuitry, and the ability to penetrate non-metallic materials, addressing spectrum scarcity and capacity challenges [4].

Several researchers have explored THz antenna designs and channel measurements to advance wireless communication. At USC, researchers used a channel sounder up to 5.5 m for THz channel analysis [5]. Shanghai Jiao Tong University and Huawei measured indoor THz channels (201–209 GHz) in various scenarios and proposed a hybrid ray-tracing-statistical model [6]. A reduced graphene-based antenna ($500 \times 500 \times 200 \mu\text{m}^3$) achieved 100 GHz bandwidth and 7.99 dB gain for compact applications [7], while another graphene patch on polyimide (725–775 GHz) offered 50 GHz bandwidth and 5.09 dBi gain [8]. A dual-band array antenna ($2920 \times 1055 \times 10 \mu\text{m}^3$) operated at 0.714 and 0.7412 THz with gains of ~ 5 dB [9]. Ahmad et al. proposed a photonic crystal-based patch antenna with 28.97 GHz bandwidth and 9.45 dB gain at 0.63 THz [10]. Jha and Singh analyzed a rectangular patch on dual-layer substrate achieving 73.12% efficiency at 542 GHz [11], while another design using photonic crystals reached 83.67% efficiency and 100 GHz bandwidth at 0.8883 THz [12]. Further designs include a polyimide-based antenna (0.665–0.696 THz) with -81.18 dB return loss and 5.22 dB gain for imaging and defense [13], and a trapezoidal antenna ($950 \times 950 \times 185 \mu\text{m}^3$) covering 0.88–1.62 THz for high-speed THz use [14]. Rectangular antenna ($\sim 1000 \times 1000 \times 200 \mu\text{m}^3$) operating at 0.7–0.95 THz achieved up to 8.248 dB gain and 88.3% efficiency using electromagnetic crystal substrates [15–16]. Overall, multi-band miniaturized THz antennas improve bandwidth, data rates, and cost-effectiveness, making them ideal for compact, short-range ultra-broadband THz systems.

To address the aforementioned limitations, this paper proposes a multifunctional triple-wideband THz antenna designed for future high-speed, short-range indoor wireless communication. The antenna operates across 0.27–0.35 THz, 0.89–1.35 THz, and 1.79–3.48 THz frequency bands. Built on a Rogers RT5880 substrate with compact dimensions of $300 \times 300 \times 15 \mu\text{m}^3$, the design ensures high radiation efficiency. Section II details the geometry of the double-overlapped E-shaped antenna, while Section III presents simulation results including S-parameters, radiation patterns, gain, directivity, and efficiency, along with comparative analysis. Section IV concludes the study.

II. THz ANTENNA DESIGN AND EVOLUTION

Initially, the antenna design begins with two overlapped elliptical patches, each having an outer radius of major axis of $103 \mu\text{m}$ and an outer radius of minor axis of $88 \mu\text{m}$. A microstrip line feeder of length $d = 73.21 \mu\text{m}$ is connected at the junction of these elliptical patches to efficiently excite the antenna. This configuration, referred to as Stage 1, employs a full metallic ground plane. In Stage 2, for enhanced performance and bandwidth optimization, the elliptical patches are innovatively modified into double-overlapped E-shaped structures by etching slots into the elliptical geometries. This modification significantly improves impedance matching and bandwidth response. To achieve omnidirectional radiation characteristics, Stage 3 replaces the full ground plane with a narrow partial ground plane of length $r = 62 \mu\text{m}$. This alteration not only improves the radiation pattern but also contributes to antenna miniaturization. Finally, in Stage 4, two L-shaped parasitic elements are incorporated as non-central corrugations within the narrow partial ground. These parasitic structures help further fine-tune the radiation properties and achieve a miniaturized, triple-wideband THz antenna, ideal for high-speed indoor wireless communication applications. The complete evolutionary design process of the proposed double-overlapped E-shaped THz antenna is illustrated in Fig. 1, while the final antenna structure, including front and back views with precise dimensions, is presented in Fig. 2. The antenna structure comprises a partially rectangular slotted ground plane, a microstrip line-fed dual-overlapped e-shaped patch that occupies the entire surface of the substrate, and two parasitic elements positioned at the non-center of the edge-cut e-shaped structure. The outer radii of the patch along the major and minor axes are $103 \mu\text{m}$ and $88 \mu\text{m}$, respectively. A copper patch ($\sigma = 5.8 \times 10^7 \text{ S/m}$) serves as the radiating element, mounted on a $300 \mu\text{m}$ square substrate. The opposite side has a slotted copper ground plane measuring $138.5 \times 128.5 \mu\text{m}^2$. The thickness of both the radiating element and the ground plane is $2 \mu\text{m}$, while the dielectric substrate maintains a thickness of $15 \mu\text{m}$. Rogers RT5880 was selected for its excellent radiation efficiency, particularly in high-frequency applications. The proposed triple-wideband THz antenna operates over three frequency ranges: 0.27–0.35 THz, 0.89–1.35 THz, and 1.79–3.48 THz. The detailed dimensions of the antenna are provided in Table I.

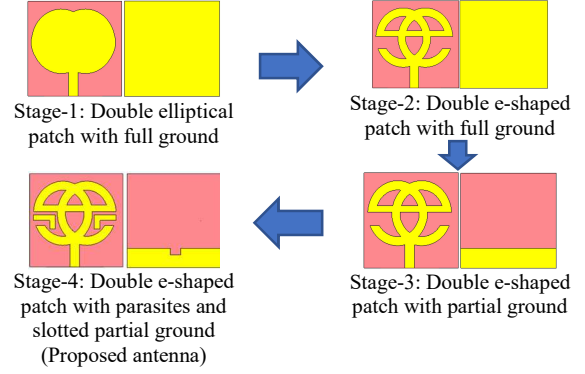


Fig. 1. Design evolution of the double-overlapped e-shaped THz antenna

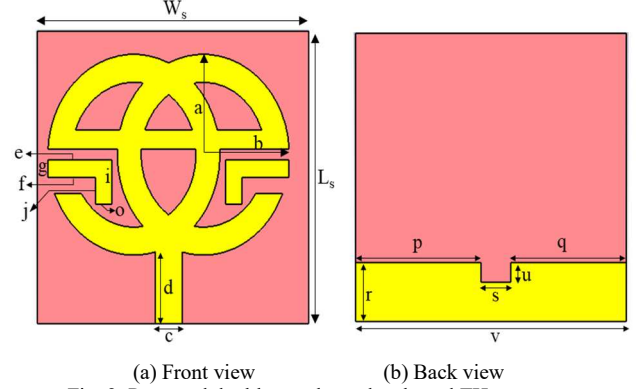


Fig. 2. Proposed double-overlapped e-shaped THz antenna

TABLE I PHYSICAL DIMENSIONS OF THE THz ANTENNA

Symbol	Value(μm)	Symbol	Value(μm)	Symbol	Value(μm)
W_s / L_s	300	e	70	o	18
a	103	f	52	p	138.50
b	88	g	18	q	128.50
c	30	i	46	r	62
d	73.21	j	28	s	33

III. RESULTS AND PERFORMANCE ANALYSIS

The scattering parameters for all four stages in the evolution of the proposed triple-wideband THz antenna, as well as for the final double-overlapped E-shaped THz antenna, are illustrated in Figs. 3(a) and 3(b), respectively. The progressive enhancement in performance resulting from various design modifications—such as slotting, implementing a partial ground plane, introducing L-shaped parasitic elements, and adding corrugations—can be clearly observed in Fig. 3(b). Among the four stages, Stage 4 exhibits the most optimized scattering characteristics, achieving the desired triple-wideband THz performance, making it suitable for high-speed indoor wireless communication systems. As shown in Fig. 3(b), the proposed antenna resonates in three distinct frequency bands: 0.27–0.35 THz, 0.89–1.35 THz, 1.79–3.48 THz. It exhibits deep reflection coefficient dips (S_{11}) at multiple resonance points: 0.31 THz (-15 dB), 1.09 THz (-32.82 dB), 1.21 THz (-41.44 dB), 1.97 THz (-22.67 dB) and 3.17 THz (-23.03 dB). These values indicate excellent impedance matching between the antenna and the transmission line/source across all three bands, resulting in minimal power reflection and thereby high radiation

efficiency. A low reflection coefficient signifies that the majority of the input power is effectively radiated or received by the antenna, with minimal loss due to reflection. The antenna successfully covers three wide frequency bandwidths: 90 GHz (0.27–0.35 THz), 460 GHz (0.89–1.35 THz) and 1690 GHz (1.79–3.48 THz). These characteristics make the proposed design highly suitable for next-generation high-speed THz wireless indoor communication applications

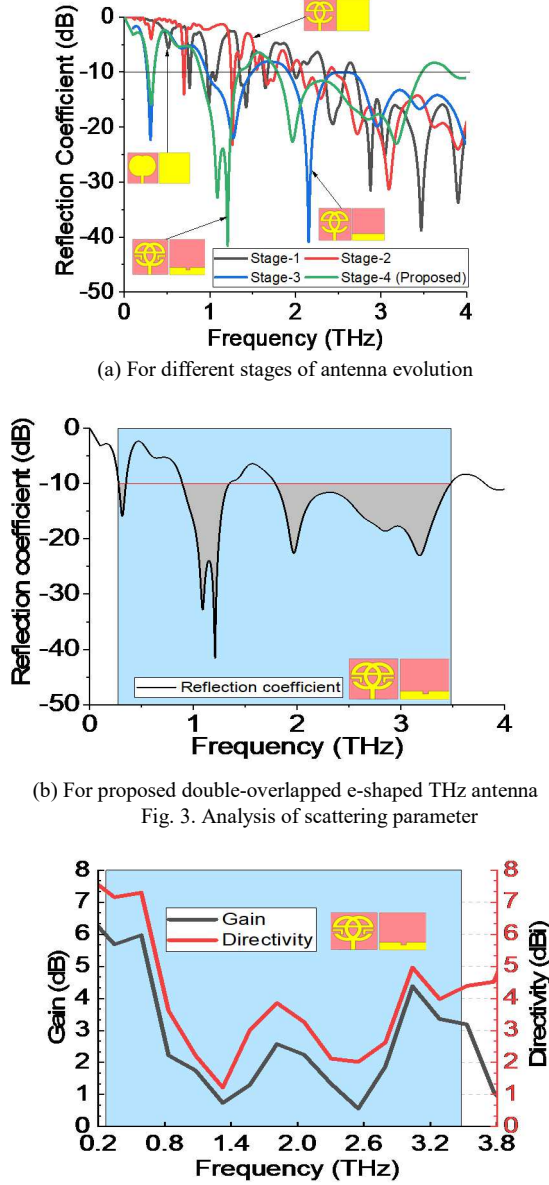


Fig. 4. Gain and directivity of the overlapped e-shaped THz antenna.

The gain and directivity of the double-overlapped e-shaped THz antenna in the triple operating bands are shown in Fig. 4. At resonant frequencies of 0.31, 1.09, 1.21, 1.97, and 3.17 THz, the antenna achieves gains of 5.89, 1.71, 1.22, 2.37, and 3.84 dB, respectively, with an overall gain range of 0.57–5.89 dB. Maximum gain of 5.89 dB is observed at 0.31 THz, while gains of 2.00 dB at 1.08 THz and 4.39 dB at 3.04 THz are noted for the other bands. The directivity at the same frequencies is 7.16, 2.17, 1.68, 3.48, and 4.44 dBi, with peak

values of 7.15 dBi at 0.31 THz and 4.39 dBi at 3.04 THz. The polar radiation patterns at 0.31 THz, 1.09 THz, 1.21 THz, 1.97 THz, and 3.17 THz are shown in Figs. 5(a-e). At 0.31 THz, side lobe levels are -1.9 dB ($\phi = 0^\circ$) for the electric field and -1.0 dB ($\phi = 90^\circ$) for the magnetic field. At 1.09 THz, main lobes are at 93° ($\phi = 0^\circ$) and 155° ($\phi = 90^\circ$), with HPBW of 141.4° ($\phi = 0^\circ$) for E and 77.1° ($\phi = 90^\circ$) for H. The electric and magnetic field magnitudes are 16 dBV/m and -35.5 dBA/m, with side lobe levels of -1.8 dB at both $\phi = 0^\circ$ and 90° . At 1.21 THz, the main lobe magnitudes are 18–21 dBV/m (E) and -33.33 dBA/m (H), with lobes at 28° ($\phi = 0^\circ$) and 30° ($\phi = 90^\circ$), and side lobes of -2.4 dB ($\phi = 0^\circ$) and -1.3 dB ($\phi = 90^\circ$). The 3 dB angular widths are 68.8° (E) and 24.4° (H). At 1.97 THz, the main lobes and HPBWs are at 112° and 68.8° ($\phi = 0^\circ$), and 153° and 24.4° ($\phi = 90^\circ$) for the E and H fields, with a minimum side lobe magnitude of -3.2 dB. At 3.17 THz, the main lobes are directed at 98° on average in both $\phi = 0^\circ$ and $\phi = 90^\circ$ planes, with 3 dB beamwidths of 59.5° ($\phi = 90^\circ$) and 31.7° ($\phi = 0^\circ$). The electric field varies from 17.6 to 19.3 dBV/m, and the magnetic field ranges from -32 to -34 dBA/m in both planes. The designed e-shaped triple band THz antenna shows maximum radiation efficiency of 90%, 91.1% and 92.8% for 0.27–0.35 THz, 0.89–1.35 THz and 1.79–3.48 THz bands as shown in Fig. 6. It exhibits VSWR values ranging from 1.02–1.16 over the whole frequency range. The VSWR value stays near to unity throughout the triple operational bandwidths, confirming excellent impedance matching. Table II clearly highlights the key strengths of the proposed THz design. Due to its ultra-compact size, triple-wideband THz operation, and high efficiency with excellent VSWR and *good* gain performance, the double-overlapped e-shaped antenna with parasitic elements emerges as a strong candidate for high-speed THz wireless indoor communication systems. Its ability to deliver reliable, high-throughput connectivity in indoor environments makes it a highly promising solution for next-generation THz applications.

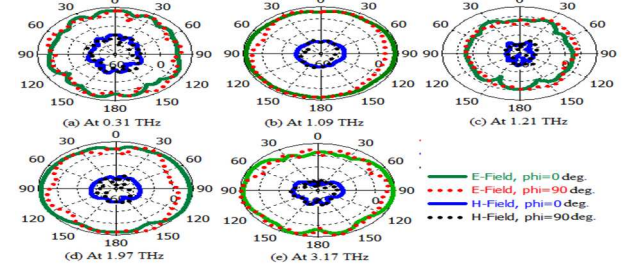


Fig. 5. E and H-field of the double-overlapped e-shaped THz antenna.

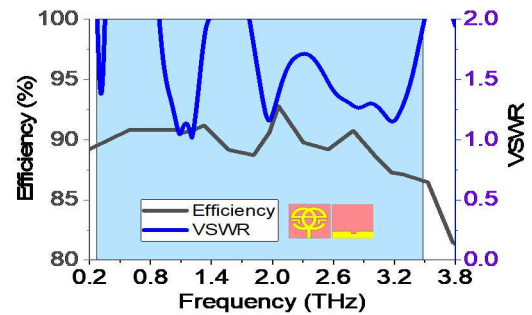


Fig. 6. Efficiency and VSWR of the e-shaped THz antenna

V. CONCLUSION AND DISCUSSION

This research presents a highly efficient, triple-wideband miniaturized THz antenna designed for emerging indoor high-speed THz applications. The antenna design begins with the modification of two overlapped elliptical patches, which are reshaped into a double-overlapped e-shaped structure. To further enhance performance, two L-pattern parasitic elements and a non-central corrugation on a narrow partial ground plane are introduced—culminating in a design that achieves triple-wideband impedance bandwidths ($S_{11} \leq -10$ dB) of 80 GHz (0.27–0.35 THz), 460 GHz (0.89–1.35 THz), and 1690 GHz

(1.79–3.48 THz). The antenna volume is only 1.35 mm³. Within the operating bands, the antenna exhibits high radiation efficiencies of 90%, 91.1%, and 92.8%, respectively. It also maintains a VSWR below 2, peak gain of 5.89 dB, maximum directivity of 7.15 dBi, and low side lobe levels with well-defined radiation patterns. Due to its ultra-compact size, broad bandwidths, and excellent electromagnetic performance, the proposed THz antenna is a strong candidate for short-range high-speed THz wireless indoor communication, as well as applications in sensing, detection, imaging, and beyond.

TABLE II: COMPARISON TABLE

Ref.	Size (L×W×h) μm ³	Substrate material	Patch material	Frequency range (THz)	BW (GHz)	Reflection coefficient (dB)	Max. Gain (dB)	Max Efficiency (%)	Application
[7]	500×500×200	Arlon (AR-100)	Graphene	≈1.2-1.3	100	-24	7.99	91.1	Size-limited communication
[8]	208.98×433.2×20	Polyimide	Graphene	0.725-0.775	50	≈-36	5.09	86.58	THz communication
[9]	2920×1055×10	Polyimide	PEC	≈0.711–0.716, ≈0.739–0.742	4.71, 3.13	-23.57, -20	12.5, 11.23	99.76, 96.6	Space communication
[10]	700×660×192	Polyimide	NA	≈0.61-0.65	40	30.66	9.45	82.6-90.6	THz communication
[11]	400×410×400	PTFE+ Arlon (AR-600)	Copper	0.500-0.625	125	≈-69	9.585	73.12	THz communication
[12]	1402×949.613×250	Homogeneous PBG substrate	Graphene	0.84-0.94	100	-57.052	14.69	83.67	THz communication
[13]	208.9×422×20	Polyimide	Copper	0.65-0.69	40	-81.18	5.22	NA	Video rate imaging and homeland defense
[14]	950×950×185	Rogers RO3003	NA	0.88–1.62	740	-15.70, -27.23, -34.28, -33.6	3.8, 10.35, 8.07, 10.50	NA	High speed THz communication
[15]	1000×1000×200	RT/Duriod6006	NA	0.7-0.85	150	≈-41	3.5	55.88	Short distance communication
[16]	1000×1000×200	Electromagnetic crystal	NA	0.6-0.95	350	≈-23, ≈-22, ≈-19	4.50, 9.19, 8.25	67.02, 76.45, 88.3	Modern communication
Proposed work	300×300×15	Rogers RT 5880	Copper	0.27-0.35 0.89-1.35, 1.79-3.48	80, 460, 1690	-15, -32.82, -41.44, -22.67, -23.03	5.89 2.00, 4.39	90 91.1, 92.8	High speed THz indoor communication

*NA = Not available

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